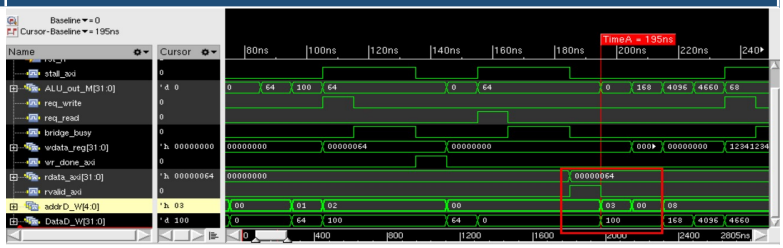
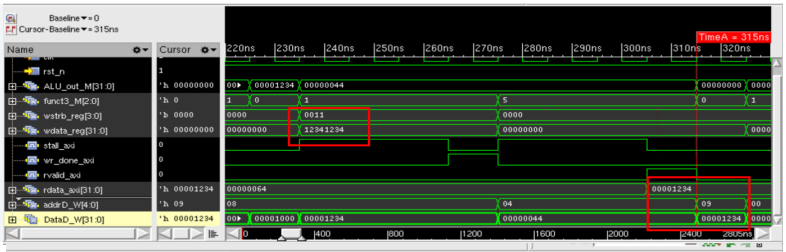
Test / Scenario	Stimulus / Method	Expected Result / Reference	Evidence
[TC1] Basic function: Basic 32-bit Store/Load	SW 100 → LW from the same address → compare	Loaded data = 100; match → PASS	
[TC2] Corner case: Half-word write and byte masking	SH 0x1234 → LHU → compare	Read data = 0x1234; correct WSTRB → PASS	
[TC3] Stress: Repeated sequential memory transactions	10 consecutive SW operations → 10 consecutive LW operations	All 10 values read back correctly; no transaction loss/deadlock → PASS	
[TC4] Hazard: Back-to-back transactions and pipeline hazards	Back-to-back SW/LW + RAW + Load-Use dependencies	Pipeline stalls correctly; forwarding resolves dependencies; final result = 700 → PASS	